

MD FAIYAZ RIZWEE

Bioinformatician | NGS Data Analysis & Genomics

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PROFESSIONAL SUMMARY

M.Sc. Bioinformatics graduate (CGPA: 9.68) with hands-on experience in end-to-end NGS data analysis, including short-read Illumina sequence alignment, SNP/INDEL variant calling, copy number variation (CNV) analysis, and RNA-Seq differential gene expression workflows on Linux/HPC environments with SLURM job scheduling. Skilled across the NGS toolchain — FastQC, HISAT2, Bowtie2, SAMtools, GATK, CNVkit, ANNOVAR, IGV — and comfortable working with FASTQ, BAM, VCF, and GFF/GTF file formats. Builds and deploys bioinformatics pipelines and web applications in Python, Django, and FastAPI, and applies machine learning to biological classification problems. Eager to extend this foundation to long-read (PacBio HiFi, Oxford Nanopore) sequencing, genome assembly, and multi-omics analysis in a production genomics lab.

TECHNICAL SKILLS

NGS & Genomics: RNA-Seq Analysis, Sequence Alignment, Variant Calling (SNP/INDEL), CNV Analysis, Differential Gene Expression, Genome Data Formats (FASTQ, BAM, VCF, GFF/GTF), SRA Toolkit, FastQC, HISAT2, Bowtie2, SAMtools, GATK, CNVkit, ANNOVAR, DESeq2, edgeR, limma-voom, IGV

Linux & HPC: Linux System Administration, Bash/Shell Scripting, High-Performance Computing (HPC) Clusters, SLURM Job Scheduling & Resource Management, Software/Environment Management, Git, GitHub, Conda, Docker

Programming & ML: Python, R (working knowledge), Scikit-learn, TensorFlow, Pandas, NumPy, Feature Engineering, Model Training & Evaluation, Classification, Regression, Data Visualization

Web & Database Development: Django, FastAPI, React.js, REST APIs, PostgreSQL, SQLite, SQLAlchemy, HTML5, CSS3, Bootstrap

RESEARCH EXPERIENCE

BRIC – National Institute of Plant Genome Research (NIPGR) *New Delhi, India*

Bioinformatics Intern → Scientific Administrative Assistant | Jan 2026 – Present

- Developing PlantPepDB 2.0, a Django-based database integrating plant peptide sequences, functional annotations, prediction tools, and interactive search utilities.
- Implementing backend modules for biological data integration, sequence retrieval, and efficient database management.
- Built a Retrieval-Augmented Generation (RAG) chatbot to answer domain-specific questions using published scientific literature as the knowledge base.
- Contributing to development and deployment of publicly accessible bioinformatics resources for plant genomics research.

NextGen Helper *Remote*

Bioinformatics Trainee | 2 Months

- Developed end-to-end NGS analysis pipelines for sequence alignment, variant calling, and copy number variation (CNV) analysis.
- Processed high-throughput sequencing datasets using HISAT2, Bowtie2, SAMtools, GATK, CNVkit, and SRA Toolkit on Linux-based systems.
- Performed SNP/INDEL identification and downstream variant interpretation across genomic datasets.

Jawaharlal Nehru University (JNU) *New Delhi, India*

Summer Research Intern | 2 Months

- Automated collection of epilepsy-associated research papers and extracted gene–disease relationships via Python-based web scraping.
- Curated and validated biological datasets to identify the top 300 candidate genes for downstream analysis.
- Constructed biological interaction networks in Cytoscape and performed network topology analysis to identify hub biomarkers.

PROJECTS

RNA-Seq Analysis Pipeline *SRA Toolkit • FastQC • HISAT2 • SAMtools • DESeq2*

- Downloaded raw sequencing reads from NCBI SRA using SRA Toolkit and assessed read quality with FastQC.

- Aligned reads to the reference genome using HISAT2, chosen for its splice-aware alignment, correctly mapping reads across exon–exon junctions in transcript data.
- Converted SAM to BAM, then sorted and indexed alignments using SAMtools for downstream processing.
- Quantified transcript abundance and performed differential gene expression analysis using DESeq2, followed by visualization and biological interpretation of DEGs.

Variant Calling Pipeline *SRA Toolkit • FastQC • Bowtie2 • GATK • CNVkit • ANNOVAR*

- Downloaded raw sequencing data from SRA and assessed read quality using FastQC.
- Aligned reads to the reference genome using Bowtie2, chosen over a splice-aware aligner since variant calling works on genomic DNA with no splice junctions, giving cleaner, more consistent mapping.
- Converted SAM to BAM, sorted and indexed with SAMtools, then applied GATK (Base Quality Score Recalibration and filtering) to call high-confidence SNPs and Indels.
- Integrated CNVkit for copy number/structural variant profiling and ANNOVAR for functional annotation of clinically relevant mutations.

Multi-Class Plant Peptide Family Predictor

Python • Scikit-learn

- Built an end-to-end ML platform for multiclass classification of plant peptide families directly from protein sequences.
- Curated peptide datasets engineered sequence-derived features and trained multiple models to identify the optimal predictor.
- Deployed the ensemble model as a web app integrated into PlantPepDB 2.0, plus a standalone CLI for high-throughput batch prediction.

GitHub: [multi-class-plant-peptide-predictor](#)

Gene2Therapy *React.js • FastAPI • pyDESeq2*

- Developed a full-stack platform for differential gene expression (DEG) analysis using React.js, FastAPI, and pyDESeq2.
- Automated downstream analyses: volcano plots, KEGG pathway enrichment, disease association (OpenTargets), and interactive network visualization.
- Integrated NCBI, KEGG, and OpenTargets into a unified analytical workflow for transcriptomic data.

Live Demo: [gene2therapy](#)

DrugLab — CLI Platform for Computational Drug Discovery *Python • AutoDock Vina • Admet AI*

- Built a lightweight, GPU-free, cloud-free CLI platform automating the full CADD workflow — protein/ligand preparation, binding-pocket detection, and AutoDock Vina docking.
- Engineered multi-criteria lead selection combining drug-likeness filters (Lipinski, Veber, PAINS, Brenk) with ADMET screening, plus a lead-optimization module for analogue generation and docking-validated candidate ranking.
- Designed a modular, fully reproducible pipeline with single-command execution, resumable stages, and config snapshots for every run.

GitHub: [druglab](#)

EDUCATION

Jamia Millia Islamia — M.Sc. Bioinformatics

2024 – 2026

CGPA: 9.68 | New Delhi, India

B.N. College, Patna University — B.Sc. Biotechnology

2019 – 2022

CGPA: 7.79 | Patna, India

PUBLICATION

Rizwee MF, Ali AM, Shah SN, Shamim T, Khan A, et al. “Patient-centric care and digital health tools.” *Advances in Protein Chemistry and Structural Biology*, Elsevier; 2025. DOI: 10.1016/bs.pbr.2025.08.003. (Co-author).